Cloud Computing and DevOps: A Combination that can Transform an Organisation

Cloud computing is becoming ubiquitous in today's competitive and dynamic business world. When combined with the DevOps culture, enterprises and developers stand to gain much in terms of quicker runtimes, scaling on demand, consistency of code, and better performance of software and applications.

Organisations are striving to gain a competitive advantage in an extremely dynamic business world. Cloud computing is a disruptive innovation that has helped to transform organisations in a way that was never thought of earlier. 'As a service' models help organisations to become more agile and swift in dynamic business scenarios.

One of the most popular images around the time cloud computing was gaining popularity was that of 'the blind men and the elephant'. Everyone who has worked on a specific aspect or characteristic of cloud computing has defined the cloud as per one's own understanding. Let us see how NIST (National Institute of Standards and Technology) defines cloud computing. According to NIST, "Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics, three service models and four deployment models."

Cloud service models

Cloud service models specify how services are made available to cloud consumers or end users.

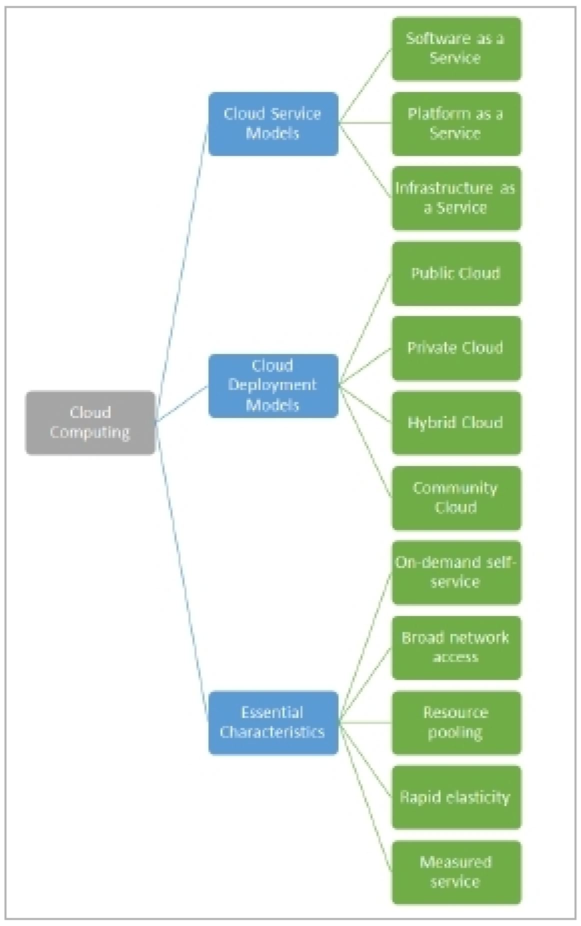

Figure 1: Cloud computing. *Source:* NIST

- **Software as a Service:** We use SaaS day in day out. The software or application is available as a service and hence we don't need to manage or install it. The application is provided by the application owner, and all the responsibilities of managing the updates of that application are of the owner and not of the consumer. Google Docs is one example of SaaS.
- **Platform as a Service:** PaaS is a model where a platform or a runtime environment is provided by the service provider. It is the responsibility of the cloud consumer to upload the application and manage an up-to-date version of the application or software. Resources are managed by the cloud service provider. Backup and all other responsibilities also lie with the service provider. PaaS offerings provide features to maintain high availability and fault tolerance, but these need to be configured by cloud consumers. Amazon Elastic Beanstalk and Microsoft Azure App Services are two popular examples